

LIN'S HOT-COLD MAXIMUM-DETERMINANT CORRESPONDENCE AT ORDER TEN

ABSTRACT. A *cold* matrix is a skew-symmetric matrix with zero diagonal and off-diagonal entries ± 1 ; the *hot* matrix attached to it is $H = C + I$. Writing $f_k(n)$ and $g_k(n)$ for the largest determinants attained in the two families of order n , Lin asked, for $n \equiv 2 \pmod{4}$, whether $\det H = g_k(n)$ holds precisely when $\det(H - I) = f_k(n)$. We settle the cell $n = 10$ by exhaustion. Bordering reduces the 2^{45} labelled order-10 cold matrices to the 191,536 isomorphism classes of 9-vertex tournaments, and a census of a covering by $6880 \cdot 2^8 = 1,761,280$ bordered matrices gives $f_k(10) = 33489 = 183^2$ and $g_k(10) = 64000 = 125 \cdot 2^9$, with *exactly one* switching-and-permutation class attaining either value. That the 6880-tournament list carrying the covering is complete and has no repeated class is not imported: it is certified inside the verification by an exact orbit count against the 2^{28} labelled 8-vertex tournaments. Lin's equivalence therefore holds at $n = 10$ in both directions, quantified over all maximisers. The maximiser is exhibited, and its two determinants are read off one characteristic polynomial by hand. Both values are already reported in the literature, and $f_k(10)$ is proved there.

1. THE QUESTION

The source is a post on Peter Cameron's blog [3], catalogued as Cameron's Problem 104. The definitions are Cameron's:

"We define a hot matrix to be a square matrix with entries ± 1 , with $+1$ on the diagonal and off-diagonal entries skew-symmetric" (sic) "; a cold matrix is a skew-symmetric matrix with 0 on the diagonal and all off-diagonal elements ± 1 . There is an obvious bijection between cold and hot matrices, given by $H = C + I$."

and so is the question:

"Dennis Lin asked whether it is true that, for n congruent to $2 \pmod{4}$, if H is a hot matrix with maximum determinant, then $H - I$ is a cold matrix with maximum determinant. Perhaps it goes the other way as well."

Write $\mathcal{F}_k(n)$ for the cold matrices of order n and $\mathcal{G}_k(n) = \{C + I : C \in \mathcal{F}_k(n)\}$ for the hot ones, and put

$$f_k(n) = \max_{C \in \mathcal{F}_k(n)} \det C, \quad g_k(n) = \max_{H \in \mathcal{G}_k(n)} \det H.$$

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Odd orders are excluded because a skew-symmetric matrix of odd order is singular, and orders divisible by 4 are settled by Hadamard's bound whenever a skew-Hadamard matrix exists; "What happens if n is not a multiple of 4? This was Lin's question" [3]. The same post records the evidence available for the first such order, and warns that the bijection $C \mapsto C + I$ "does not induce a monotone map on their determinants":

"There are too many matrices for $n = 10$ for an exhaustive search; and in lieu of doing anything clever, I simply looked at a few million random matrices ... (The best I found were 64000 for hot and 33489 for cold.)"

Armario and Frau restate the question as an explicit equivalence in a refereed journal [2, §1]: "it asks to decide whether or not the following equivalence holds: M has maximum determinant among the matrices in $F_k(n)$ if and only if $M + I$ has maximum determinant among the matrices in $G_k(n)$ ". That is the statement we prove at $n = 10$. Their Theorem 2.6 proves the equivalence when $f_k(n) = (2n - 3)(n - 3)^{(n-2)/2}$, which by their Proposition 2.4 requires $2n - 3$ to be a perfect square, so at $n = 10$, where $2n - 3 = 17$, it does not apply.

2. THE RESULT

Theorem 1. $f_k(10) = 33489 = 183^2$ and $g_k(10) = 64000 = 125 \cdot 2^9$. Moreover exactly one switching-and-permutation class of order-10 cold matrices attains $f_k(10)$, exactly one attains $g_k(10)$, and the two classes are the same one, namely that of $C = H - I$ for the matrix H displayed in (1) below. Consequently, for $n = 10$, a hot matrix has maximum determinant if and only if the cold matrix $H - I$ has maximum determinant.

The witness, given as the hot matrix $H = C + I$: rows run top to bottom, + is +1 and - is -1; H is hot, i.e. its diagonal is +1 and $h_{ij} = -h_{ji}$ off the diagonal, and $C = H - I$ is cold.

$$(1) \quad \begin{array}{c} ++++++++ \\ -+++++- \\ -++++-+ \\ --+++- \\ --+++- \\ --+++- \\ --+++- \\ -++-++ \\ -++-++ \\ -++-++ \\ -++-++ \\ -++-++ \end{array}$$

That this matrix attains both values is a hand computation. The characteristic polynomial of C is

$$(2) \quad \det(xI - C) = x^{10} + 45x^8 + 770x^6 + 6210x^4 + 23485x^2 + 33489.$$

Its constant term is $\det C = 33489 = 183^2$, and since the degree is even, $\det H = \det(I + C)$ is the value of (2) at $x = -1$, that is

$$1 + 45 + 770 + 6210 + 23485 + 33489 = 64000 = 125 \cdot 2^9.$$

The coefficient 45 is forced and needs no computation: $\text{tr}(C^\top C) = 90$ because all 90 off-diagonal entries have modulus 1, and that trace is twice the x^8 coefficient. The sorted out-degree sequence of the tournament $\{i \rightarrow j : h_{ij} = +1\}$ is 3, 3, 4, 4, 4, 4, 4, 5, 5, 9; as an undirected graph on $\{0, \dots, 9\}$ with an edge $\{i, j\}$, $i < j$, exactly when $h_{ij} = +1$, the same object is the 31-edge graph with `graph6` string `I~1krMig`.

3. THE PROOF

Only the maximality and the uniqueness remain, and they come from four lemmas and a finite census.

Lemma 2 (invariance). *Let C be cold of order n , let D be diagonal with entries ± 1 and let P be a permutation matrix. Then DCD and $PCP^\top$ are cold, and all three of $\det C$, $\det(I + C)$ and the polynomial $\det(xI - C)$ are unchanged.*

Proof. Both maps preserve the zero diagonal and the moduli of the entries, and skewness is preserved because D and P are real. Since $D = D^{-1}$ and $P^\top = P^{-1}$, both maps are similarities, so they fix the characteristic polynomial; and $I + DCD = D(I + C)D$, $I + PCP^\top = P(I + C)P^\top$, so they fix $\det(I + C)$ as well. $\square$

Call C, C' *equivalent* when $C' = PDCDP^\top$ for some such D, P ; by Lemma 2 both determinants are invariants of the equivalence class. For a tournament T on k vertices let $A(T)$ be its Seidel matrix, $a_{ij} = +1$ if $i \rightarrow j$ and -1 if $j \rightarrow i$, and put

$$\text{bd}(T) = \begin{pmatrix} 0 & \mathbf{1}^\top \\ -\mathbf{1} & A(T) \end{pmatrix},$$

a cold matrix of order $k + 1$.

Lemma 3 (bordering). *Every cold matrix of order n is equivalent to $\text{bd}(T)$ for some tournament T on $n - 1$ vertices. If $T \cong T'$ then $\text{bd}(T)$ and $\text{bd}(T')$ are equivalent.*

Proof. Index rows by $0, 1, \dots, n - 1$ and switch by $D = \text{diag}(d_0, \dots, d_{n-1})$ with $d_0 = +1$ and $d_j = c_{0j}$ for $j \geq 1$. The $(0, j)$ entry of DCD is $d_0 c_{0j} d_j = c_{0j}^2 = +1$, so row 0 of DCD is $(0, +1, \dots, +1)$ and column 0 is its negative; the remaining block is the Seidel matrix of a tournament on $n - 1$ vertices. The second assertion is the case of Lemma 2 in which P fixes index 0: those P act on the lower block as the full symmetric group on $n - 1$ letters. $\square$

This is the standard bordering of Reid and Brown, used in the same form by Klanderman, Montee, Piotrowski, Rice and Shader [4].

Lemma 4 (covering). *Let $\mathcal{S}$ be a set of tournaments on $n-1$ vertices meeting every isomorphism class. Then $\{\text{bd}(T) : T \in \mathcal{S}\}$ meets every equivalence class of cold matrices of order n . In particular $\max_{T \in \mathcal{S}} \det \text{bd}(T) = f_k(n)$ and $\max_{T \in \mathcal{S}} \det(I + \text{bd}(T)) = g_k(n)$, and if all the maximisers in $\mathcal{S}$ are isomorphic tournaments then exactly one equivalence class of cold matrices attains the corresponding maximum.*

Proof. Immediate from Lemma 3, since each class contains some $\text{bd}(T)$ and $\mathcal{S}$ contains a tournament isomorphic to that T . For the last clause: every class attaining the maximum contains some $\text{bd}(T)$ with $T \in \mathcal{S}$ maximal, and isomorphic T give equivalent matrices by Lemma 3, so all such classes coincide. $\square$

This is what makes Cameron's "too many matrices" finite. At $n = 10$ there are $2^{45} = 35,184,372,088,832$ labelled cold matrices, but at most $A000568(9) = 191,536$ equivalence classes.

The covering used below is built from a list of 8-vertex tournaments, so Lemma 4 is only as good as the guarantee that the list meets every isomorphism class. That guarantee is not taken from the literature: it is certified by the program, and the certificate is the following one.

Lemma 5 (the class list). *For $k \leq 8$ let R_k be the list obtained from $R_1 = \{K_1\}$ by forming all 2^{k-1} one-vertex extensions of every member of R_{k-1} and keeping one tournament per value of the canonical code*

$$\text{cd}(T) = \max_{\sigma} (\text{upper-triangle code of } \sigma T),$$

where σ runs over the relabellings that order the out-degrees of T non-decreasingly. Then $|R_8| = 6880$, the members of R_8 are pairwise non-isomorphic, and every 8-vertex tournament is isomorphic to exactly one of them.

Proof. Two facts, each verified computationally as stated.

(i) *Nothing is lost.* $\text{cd}(T)$ is by construction the upper-triangle code of an *explicit* relabelling of T , so two tournaments with equal code are both isomorphic to the tournament decoded from that code, hence to each other; the program checks for each of the 6880 members that the maximising relabelling reproduces the code and that decoding it returns that relabelling. Therefore the deduplication discards a tournament only in favour of an isomorphic one. Since deleting a vertex from a k -vertex tournament leaves a $(k-1)$ -vertex one, and an isomorphism of the smaller tournaments extends to the larger, every class at level k occurs among the one-vertex extensions of R_{k-1} ; by induction R_8 meets every class.

(ii) *Nothing is repeated.* A relabelling permutes out-degrees, so every automorphism fixes each vertex's out-degree and $|\text{Aut}(T)|$ may be counted over the out-degree-preserving permutations alone; the program checks that this count agrees with the count over all $k!$ relabellings for all 532 members

of $R_1, \dots, R_7$. With $|\text{Aut}|$ in hand,

$$\sum_{T \in R_k} \frac{k!}{|\text{Aut}(T)|} = 2^{k(k-1)/2} \quad (2 \leq k \leq 8),$$

verified level by level, the right-hand side being the number of *all* labelled k -vertex tournaments, 268,435,456 at $k = 8$. By (i) the orbits of the members of R_k already cover those $2^{k(k-1)/2}$ tournaments, so a repeated class would make the left-hand side exceed the right; equality forces the members to be pairwise non-isomorphic. In particular $|R_8| = 6880 = A000568(8)$ is the number of classes, not merely an upper bound for it. $\square$

Independently, the canonical code is verified to be a complete isomorphism invariant by exhaustion at two orders: over all 1024 labelled 5-vertex tournaments each code class is checked to be exactly one full 120-relabelling orbit, and over all 32,768 labelled 6-vertex tournaments the codes are checked to take exactly $56 = A000568(6)$ values with each class of size $720/|\text{Aut}|$.

Proposition 6 (the census). *Take $\mathcal{S}$ to consist of the $2^8 = 256$ one-vertex extensions of each of the 6880 members of the list R_8 of Lemma 5, so $|\mathcal{S}| = 1,761,280$; deleting a vertex from a 9-vertex tournament leaves an 8-vertex one, which by Lemma 5 is isomorphic to a member of R_8 , and that isomorphism extends, so $\mathcal{S}$ meets every isomorphism class of 9-vertex tournaments. Over $\mathcal{S}$,*

$$\max_T \det \text{bd}(T) = 33489, \quad \max_T \det(I + \text{bd}(T)) = 64000,$$

the two maxima are attained by the same nine members of $\mathcal{S}$, and those nine are pairwise isomorphic tournaments, forming the single class of (1).

Theorem 1 follows from Proposition 6 by Lemma 4, the last clause of that lemma giving both the uniqueness and the coincidence of the two argmax sets, and hence Lin's equivalence at $n = 10$ in both directions. The census is 1,761,280 pairs of exact integer 10×10 determinants, computed in exact integer arithmetic; it is carried out by `verify.py` in the accompanying material, and `verify.output.txt` is a recorded run of that program.

What else was checked. The same program reproduces the published cells $(f_k, g_k) = (9, 16)$, $(81, 160)$ and $(2401, 4096)$ at $n = 4, 6, 8$, the first two also by brute force over all 2^6 and all 2^{15} labelled cold matrices; the isomorphism-class counts 1, 1, 2, 4, 12, 56, 456, 6880 on 1 to 8 vertices against A000568 [7]; the set of order-10 Seidel tournament determinants published in [4]; and the two order-reversing order-10 pairs recorded by Orrick [5], $(\det H, \det C) = (45056, 21609)$ and $(46080, 20449)$, which occur among the leaves.

4. RELATION TO THE LITERATURE, AND SCOPE

Both records are already in print. Alvarez, Armario, Frau and Gudiel write, in the first section of [1]: “For $n \leq 30$, orders 18, 22, 30 are unresolved.

The case $n = 10$ was solved by Cameron, $g_k(10) = 64000$ and $f_k(10) = 33489$, using a random search and again Lin's correspondence was confirmed." Moreover $f_k(10) = 33489$ is *proved* in print: the complete set of determinants of order-10 Seidel tournament matrices is computed in [4], and 33489 is its largest element. The census above establishes both values, the equivalence in both directions, and the uniqueness of the maximising class, independently and by exhaustion; it establishes nothing about priority, and we claim none. Of the two further items bearing on a possible earlier tabulation, the first has now been read and does not overlap: Zeng and You [6] study, for odd k , the hereditary class $\mathcal{D}_k$ of tournaments all of whose subtournaments have determinant at most k^2 , characterising $\mathcal{D}_5$ and constructing, for each odd k , a tournament of determinant k^2 in $\mathcal{D}_k \setminus \mathcal{D}_{k-2}$; no order is singled out, no maximum over an order is computed, and the hot matrix $C + I$ does not occur there. The second, Seberry and Yamada, *Hadamard Matrices* (Wiley, 2020), ch. 12, *More on Maximal Determinant Matrices*, pp. 245–269, we were unable to consult. It is a survey chapter on maximal determinants, so it may well record $f_k(10)$ or $g_k(10)$; as those two constants are in any case credited above to [1] and [4], the content claimed here is the exhaustive proof of Lin's equivalence at $n = 10$ quantified over all maximisers, together with the uniqueness of the maximising class, and not either constant.

The scope is the cell $n = 10$ only, against a question quantified over all $n \equiv 2 \pmod{4}$. Orders 18, 22 and 30 are unresolved before and after this note, exactly as [1] records, and the method here does not reach $n = 18$: it would require the isomorphism classes of 17-vertex tournaments. Nothing here bears on the unrestricted maximal-determinant problem.

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